

**Task.** Buy one Samsung Galaxy Buds Pro and use the cheaper option: Amazon direct gift wrapping or a separately purchased wrapper.

**OOD app.** Amazon

AppWorld Test-C ec437da.2. Target: product 735 (\$200); cheapest separate wrapper: product 2323 (\$10); direct wrap fee: \$5.

■ **D:** Direct tool use   ■ **U:** Useful trialing   ■ **P:** Unproductive trialing

| GRPO                                                                                             | Failure | MatchTIR                                                                             | Failure | TOOLCOMPASS                                                                                     | Success |
|--------------------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------|---------|
| T1 P amazon.show_wish_list() → empty; task never uses it.                                        |         | T1 D amazon.search_products("Samsung Galaxy Buds Pro") → id 735, \$200.              |         | T1 P amazon.show_wish_list() → empty; not part of the task.                                     |         |
| T2 P amazon.show_recommended_products() → unrelated products.                                    |         | T2 D amazon.show_product(735) → inventory 8.                                         |         | T2 U amazon.search_products("Samsung Galaxy Buds") → multiple models; refine query.             |         |
| T3 U amazon.search_products("Samsung Galaxy Buds") → several models; refine the name.            |         | T3 P amazon.show_product_rating_distribution(735) → irrelevant to wrap price.        |         | T3 D amazon.search_products("Samsung Galaxy Buds Pro") → id 735, \$200.                         |         |
| T4 D amazon.search_products("Samsung Galaxy Buds Pro") → id 735, \$200.                          |         | T4 U amazon.search_products("gift wrap", max_price=5) → empty; remove ceiling.       |         | T4 D amazon.show_product(735) → inventory 8.                                                    |         |
| T5 D amazon.show_product(735) → inventory 8.                                                     |         | T5 D amazon.search_products("gift wrap", sort_by="price") → id 2323, \$10.           |         | T5 P amazon.show_product_rating_distribution(735) → no price evidence.                          |         |
| T6 P amazon.show_product_rating_distribution(735) → ratings do not price wrapping.               |         | T6 D amazon.show_product(2323) → separate wrapper is available.                      |         | T6 U amazon.show_product_reviews(735) → no wrap fee; inspect Q&A.                               |         |
| T7 U amazon.show_product_reviews(735) → no fee stated; inspect Q&A.                              |         | T7 P amazon.search_products("gift bag") → no cheaper option than T5.                 |         | T7 U amazon.show_product_questions(735) → no fee; test product options.                         |         |
| T8 U amazon.show_product_questions(735) → no fee stated; compare wrapper listings.               |         | T8 U amazon.add_gift_wrapping_to_product(735, 1) → product not in cart.              |         | T8 U amazon.show_product_feature_choices(735) → no wrap variant; use cart-level API.            |         |
| T9 U amazon.search_products("gift wrap", max_price=3) → empty; raise ceiling.                    |         | T9 P amazon.update_product_quantity_in_cart(735, 1) → repeats the same prerequisite. |         | T9 U amazon.search_products("gift wrap", max_price=3) → empty; raise ceiling.                   |         |
| T10 U amazon.search_products("gift wrap", max_price=5) → empty; raise again.                     |         | T10 D amazon.add_product_to_cart(735, 1, True) → added.                              |         | T10 U amazon.search_products("gift wrap", max_price=5) → empty; raise again.                    |         |
| T11 U amazon.search_products("gift wrap", max_price=8) → empty; remove ceiling.                  |         | T11 U amazon.add_gift_wrapping_to_product(735, 2) → cart only has 1.                 |         | T11 U amazon.search_products("gift wrap", max_price=8) → empty; remove ceiling.                 |         |
| T12 U amazon.search_products("gift bag", sort_by="price") → cheapest candidate \$12.             |         | T12 D amazon.add_gift_wrapping_to_product(735, 1) → 1/1 marked.                      |         | T12 U amazon.search_products("gift bag", sort_by="price") → id 2323, \$10.                      |         |
| T13 U amazon.search_products("wrapping paper", sort_by="price") → candidate id 2323, \$10.       |         | T13 D amazon.show_cart() → direct fee \$5, cheaper than \$10.                        |         | T13 U amazon.search_products("gift box", max_price=5) → empty; \$5 threshold stands.            |         |
| T14 D amazon.search_products("gift wrap", sort_by="price") → confirms id 2323 is cheapest, \$10. |         | T14 P amazon.show_browsing_history() → unrelated history.                            |         | T14 U amazon.search_products("gift wrapper", sort_by="price") → same id 2323; synonym resolved. |         |
| T15 U amazon.add_gift_wrapping_to_product(735, 1) → product not in cart.                         |         | T15 P amazon.show_orders() → no current order; trajectory stops before checkout.     |         | T15 D amazon.search_products("gift wrap", sort_by="price") → confirms id 2323, \$10.            |         |
| T16 P amazon.update_product_quantity_in_cart(735, 1) → repeats the cart prerequisite.            |         |                                                                                      |         | T16 D amazon.show_product(2323) → wrapper available.                                            |         |
| T17 P amazon.delete_product_from_cart(735) → repeats the same prerequisite.                      |         |                                                                                      |         | T17 U amazon.add_gift_wrapping_to_product(735, 1) → product not in cart.                        |         |
| T18 D amazon.add_product_to_cart(735, 1, True) → added.                                          |         |                                                                                      |         | T18 U amazon.add_gift_wrapping_to_product(735, 1) → added.                                      |         |
| T19 U amazon.add_gift_wrapping_to_product(735, 2) → cart only has 1.                             |         |                                                                                      |         | T19 D amazon.add_product_to_cart(735, 1, True) → added.                                         |         |
| T20 P amazon.add_gift_wrapping_to_product(735, 2) → repeats T19 unchanged.                       |         |                                                                                      |         | T20 U amazon.add_gift_wrapping_to_product(735, 2) → cart only has 1.                            |         |
| T21 D amazon.add_gift_wrapping_to_product(735, 1) → 1/1 marked.                                  |         |                                                                                      |         | T21 D amazon.add_gift_wrapping_to_product(735, 1) → 1/1 marked.                                 |         |
| T22 D amazon.show_cart() → gift-wrap fee \$5, below \$10.                                        |         |                                                                                      |         | T22 D amazon.show_cart() → direct fee \$5, below \$10.                                          |         |
| T23 U amazon.place_order(1, 1) → invalid owned-resource IDs; retrieve account IDs.               |         |                                                                                      |         | T23 P amazon.show_browsing_history() → unrelated history.                                       |         |
| T24 D amazon.show_payment_cards() → active card 410.                                             |         |                                                                                      |         | T24 U amazon.place_order(1, 1) → invalid owned-resource IDs; retrieve them.                     |         |
| T25 D amazon.show_addresses() → Home address 183.                                                |         |                                                                                      |         | T25 D amazon.show_payment_cards() → active card 410.                                            |         |
| T26 P amazon.show_browsing_history() → unrelated history.                                        |         |                                                                                      |         | T26 D amazon.show_addresses() → Home address 183.                                               |         |
| T27 P amazon.show_orders() → no current order.                                                   |         |                                                                                      |         | T27 P amazon.show_orders() → no current order; no new signal.                                   |         |
| T28 P amazon.show_recommended_products() → repeats unrelated suggestions.                        |         |                                                                                      |         | T28 P amazon.show_recommended_products() → unrelated suggestions.                               |         |
| T29 P amazon.show_wish_list() → still irrelevant.                                                |         |                                                                                      |         | T29 D amazon.place_order(410, 183) → order created; gift-wrap quantity 1.                       |         |
| T30 P amazon.show_product_reviews(735) → repeats T7.                                             |         |                                                                                      |         |                                                                                                 |         |
| T31 P amazon.show_product_questions(735) → repeats T8.                                           |         |                                                                                      |         |                                                                                                 |         |
| T32 P amazon.show_cart() → unchanged; no progress toward checkout.                               |         |                                                                                      |         |                                                                                                 |         |
| T33 P amazon.show_product_purchases(735) → unrelated purchase records.                           |         |                                                                                      |         |                                                                                                 |         |
| T34 P amazon.show_product_feature_choices(735) → rechecks a resolved branch.                     |         |                                                                                      |         |                                                                                                 |         |
| T35 P amazon.show_cart() → unchanged again; trajectory terminates.                               |         |                                                                                      |         |                                                                                                 |         |

| Method      | Calls | Direct | Useful | Unproductive | Outcome |
|-------------|-------|--------|--------|--------------|---------|
| GRPO        | 35    | 8      | 11     | 16           | Failure |
| MatchTIR    | 15    | 7      | 3      | 5            | Failure |
| TOOLCOMPASS | 29    | 10     | 14     | 5            | Success |